DSP

Forever notes :)

HW

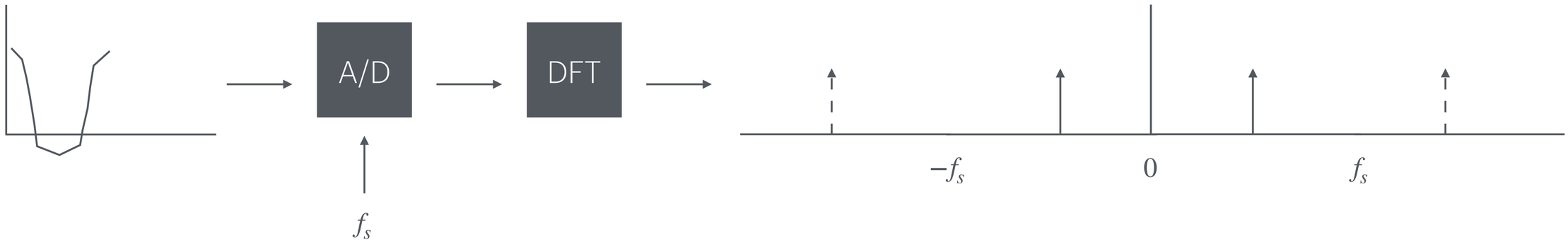
Chapter 1

$$x(t) = cos(2\pi \times f \times t)$$

Continuous Time Data

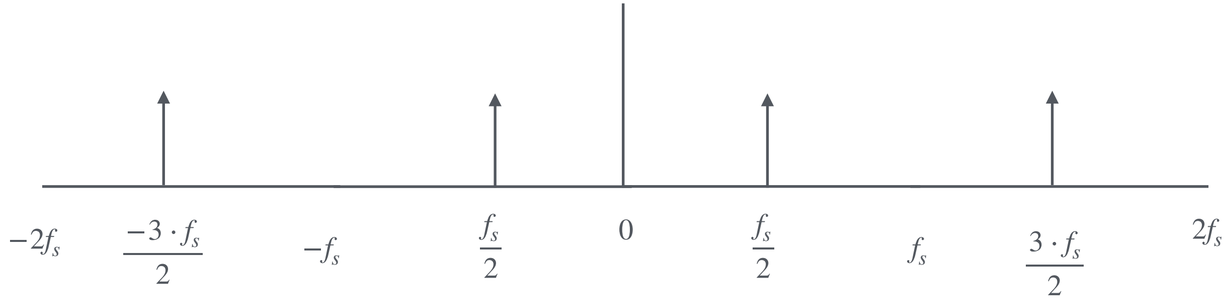
$$x(n) = cos(2\pi \times f \times n \times t_s)$$

Discrete Sampled Data

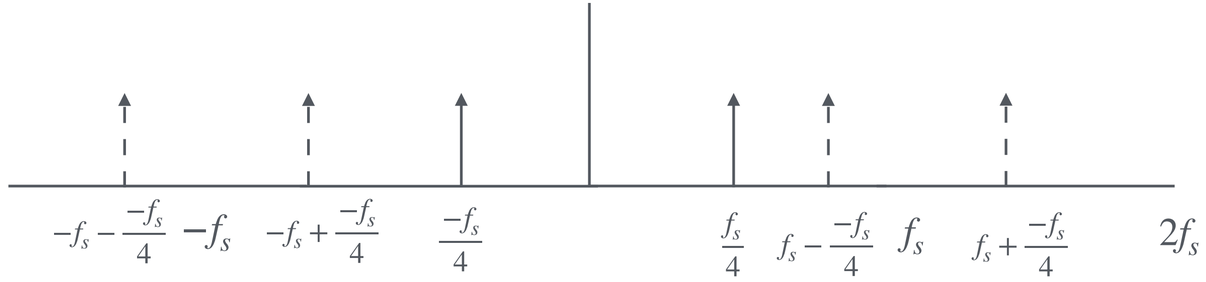


$$x(n) = cos(2\pi \times f \times n \times t_s)$$

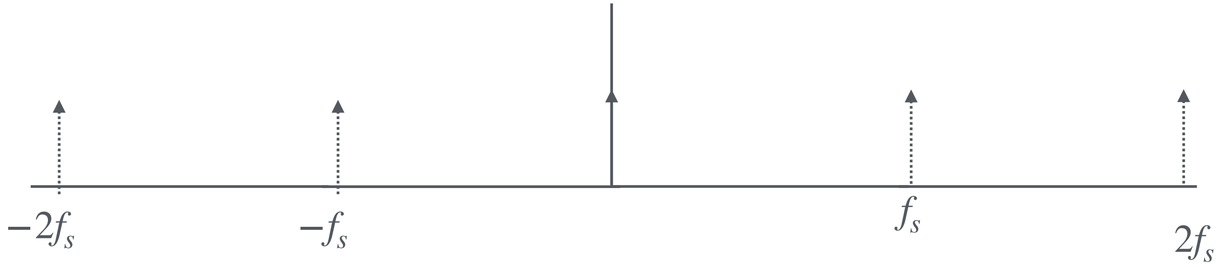
$$f = f_s/2$$



$$f = f_s/4$$



$$f = 0$$



4 Million Samples Block

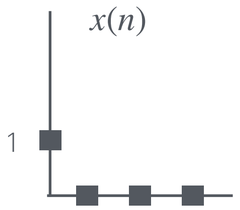
Max number cycles of analog signal able to store in memory?

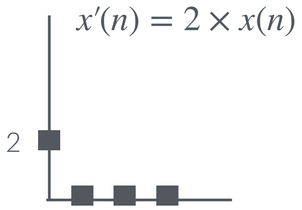
$$\frac{25\ Million\ Cycles}{Second}\ \frac{Second}{100\ MillionSamples} = \frac{1\ Cycle}{4\ Sample}$$

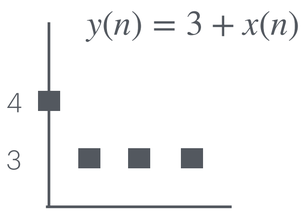
$$Conv_Rate = \frac{1\ Cycle}{4\ Sample}$$

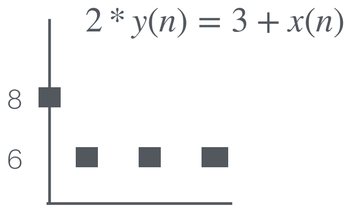
$$\frac{1\ Cycle}{4\ Sample} \times \frac{4\ Million\ Samples}{} = 1\ Million\ Cycles$$

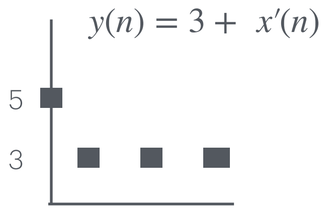
LTI System







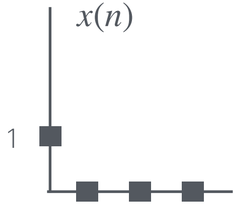




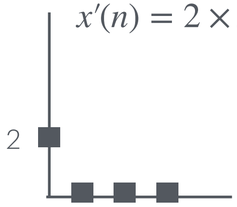
System $\neq$ LTI System

LTI System

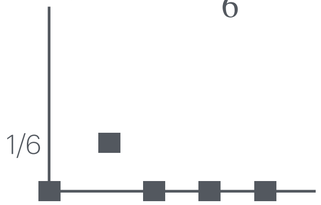
$x(n)$



$x'(n) = 2 \times x(n)$

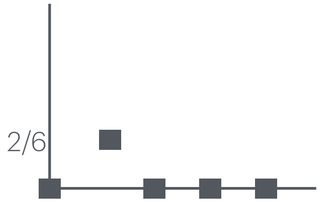


$y(n) = \dfrac{x(n-1)}{6}$



$2 * y(n) = \dfrac{x(n-1)}{6}$

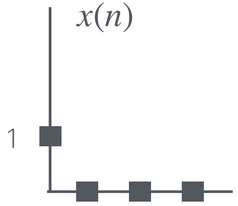
$2 * y(n) = \dfrac{x'(n-1)}{6}$



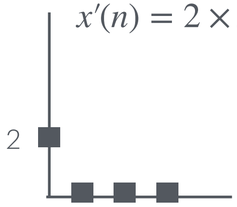
System = LTI System

LTI System

$x(n)$

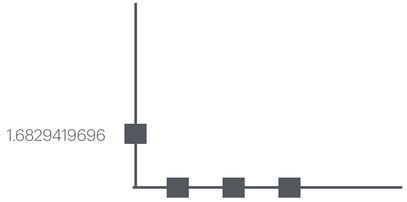


$x'(n) = 2 \times x(n)$

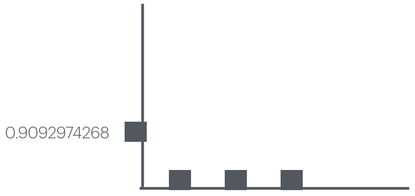


$y(n) = sin(x(n))$

$y(n) = 2 \times sin(x(n))$

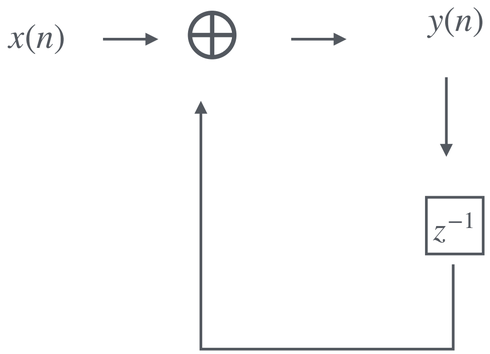


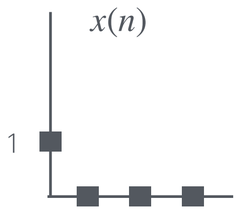
$y(n) = sin(x'(n))$

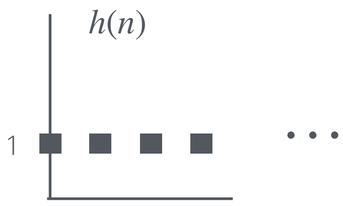


System $\neq$ LTI System

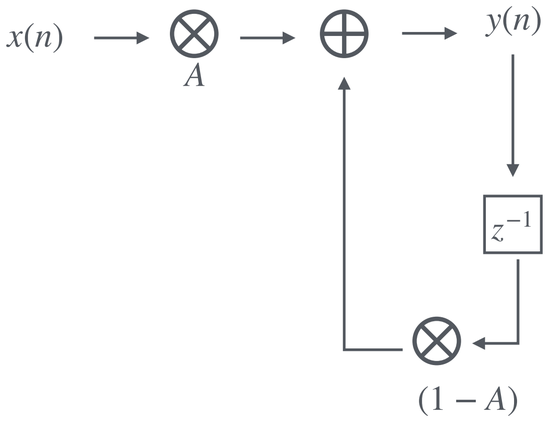
Impulse Response: Integrator

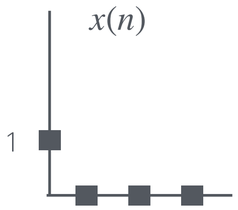


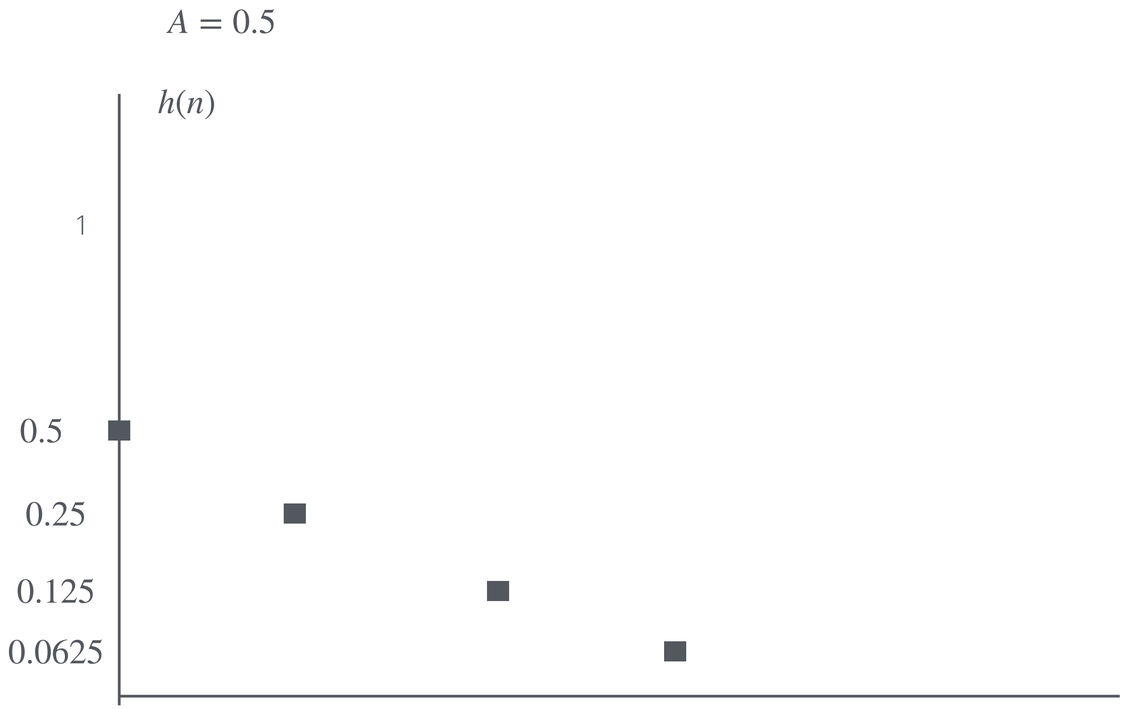




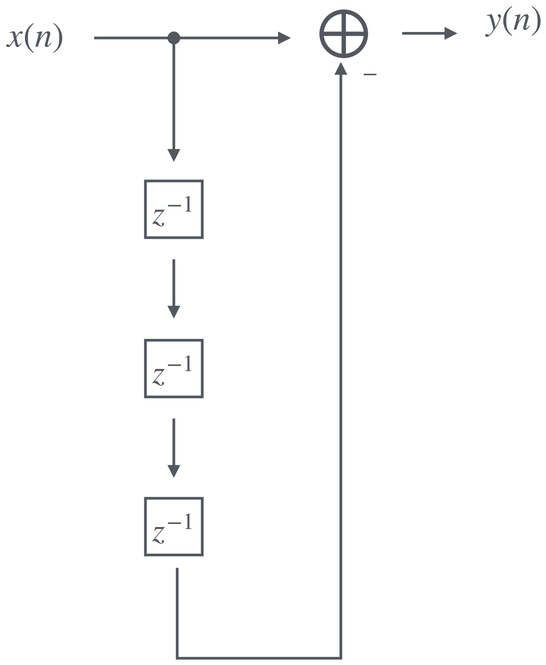
Impulse Response: Leaky Integrator

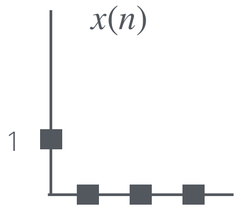


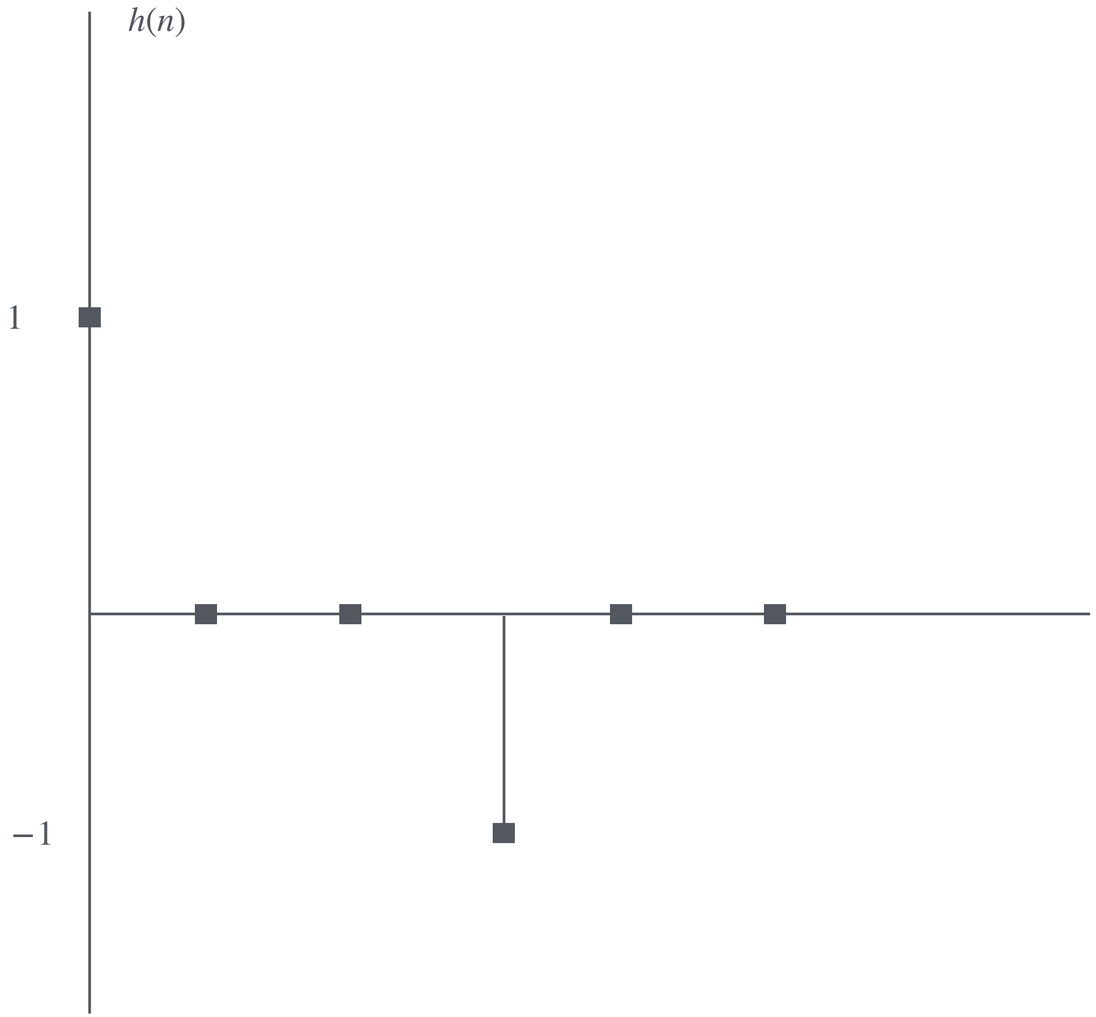




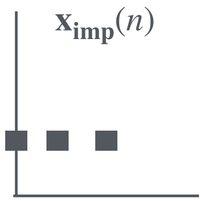
Impulse Response: Central Differentiator

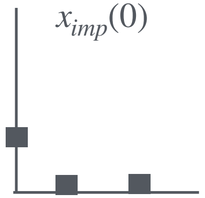


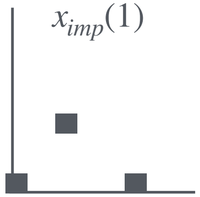


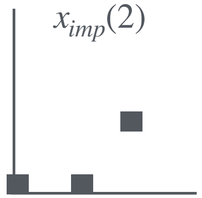


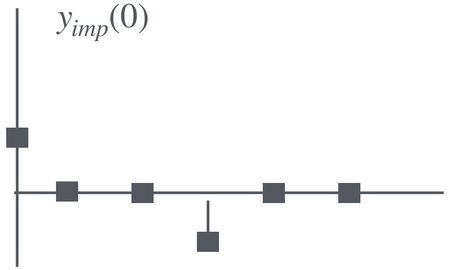
Step Response

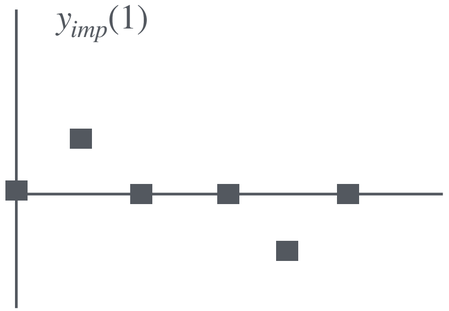


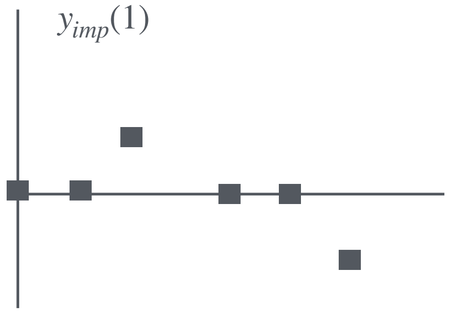






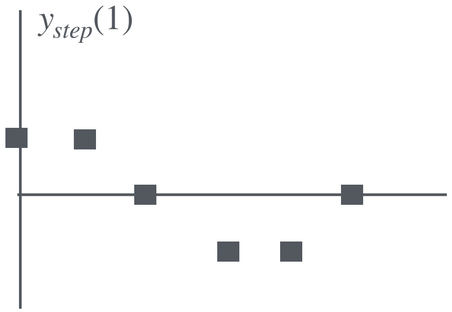
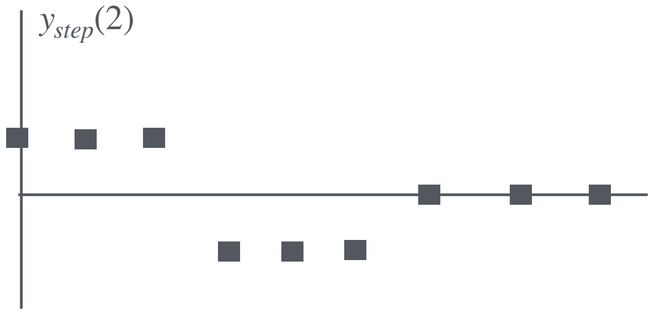






$y_{step}(0) = y_{imp}(0)$

$y_{step}(1) = y_{imp}(0) + y_{imp}(1)$

$y_{step}(2) = y_{imp}(0) + y_{imp}(1) + y_{imp}(2)$

$y_step(n)$ is accumulative sum of unit impulses